

Central Configurations (C.C.)

There is one class of general solutions to the NBP. These have been extensively studied & includes the 2BP. They exist whenever a system is placed in a C.C.

A C.C. exists whenever the accelerations acting on each body point towards the barycenter & are proportional to one body's distance to the barycenter, with the same proportionality constant.

$$m_i \ddot{\vec{r}}_i = - \frac{\partial \mathcal{U}}{\partial \vec{r}_i} = \boxed{m_i \vec{r}_i (2\sigma)} \quad \sigma \text{ is a prop. constant}$$

$i = 1, 2, \dots, N$

Defines a C.C. $\left(\sum_{i=1}^N m_i \vec{r}_i = \vec{0} \right)$

$$\frac{\partial \mathcal{I}_p}{\partial \vec{r}_i} = 2 m_i \vec{r}_i$$

c.c. condition

$$-\frac{\partial \mathcal{U}}{\partial \vec{r}_i} = \sigma \frac{\partial \mathcal{I}_p}{\partial \vec{r}_i} \Rightarrow \boxed{\frac{\partial \mathcal{U}}{\partial \vec{r}_i} + \sigma \frac{\partial \mathcal{I}_p}{\partial \vec{r}_i} = 0 \quad \forall i = 1, 2, \dots, N}$$

Only a function of relative positions.

we can simplify this w/ Euler's Thm on Homog. Funs. ★

$$\sum_{i=1}^N \left[\vec{r}_i \cdot \frac{\partial \mathcal{U}}{\partial \vec{r}_i} + \sigma \vec{r}_i \cdot \frac{\partial \mathcal{I}_p}{\partial \vec{r}_i} \right] = \boxed{0 = -\mathcal{U} + 2\sigma \mathcal{I}_p} \Rightarrow \boxed{\sigma = \frac{\mathcal{U}}{2\mathcal{I}_p}}$$

$$\sigma = \frac{\mathcal{U}}{2I_p}$$

the common prop. term must be.

σ can vary w/ time

Thus

$$\frac{\mathcal{U}}{\mathcal{J} \vec{r}_i} + \frac{\mathcal{U}}{2I_p} \frac{\mathcal{J} I_p}{\mathcal{J} \vec{r}_i} = 0 \quad i=1, \dots, N$$

Condition for
C.C.

Mult. by $(2I_p \mathcal{U})$

$$\left(2I_p \mathcal{U} \frac{\mathcal{J} \mathcal{U}}{\mathcal{J} \vec{r}_i} + \mathcal{U}^2 \frac{\mathcal{J} I_p}{\mathcal{J} \vec{r}_i} \right) = 0 =$$

$$\frac{\mathcal{J}}{\mathcal{J} \vec{r}_i} [\mathcal{U}^2 I_p] = 0 \quad i=1, 2, \dots, N$$

$\mathcal{U}^2$ is homog. of degree -2
 I_p " " " " 2

$\mathcal{U}^2 I_p$ is homog. of degree 0

$$\underline{X} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix}$$

$$\mathcal{C}(\underline{X}) = \mathcal{U}^2 I_p$$

$$\underline{X} \cdot \frac{\mathcal{J}\mathcal{C}}{\mathcal{J}\underline{X}} = 0$$

This does not mean that $\frac{\mathcal{J}\mathcal{C}}{\mathcal{J}\underline{X}} \equiv 0$!

The C.C. Condition then becomes

$$\frac{\mathcal{J}\mathcal{C}}{\mathcal{J}\underline{X}} = \vec{0}_{3N \times 1}$$

$N=2 \rightarrow$ Trivially satisfied
 $N=3 \rightarrow 5$ configurations
 $N=4 \rightarrow ?$

Compute $\mathcal{C}$ for $N=2$

$$\mathcal{U} = -\frac{G m_1 m_2}{r_{12}}$$

$$I_p = \frac{m_1 m_2}{m_1 + m_2} r_{12}^2$$

$$\mathcal{C} = \mathcal{U}^2 I_p = \frac{G^2 m_1^2 m_2^2}{r_{12}^2} \frac{m_1 m_2}{m_1 + m_2} r_{12}^2 = \frac{G^2 (m_1 m_2)^3}{m_1 + m_2}$$

$$\underline{X} = \begin{bmatrix} \vec{r}_1 \\ \vec{r}_2 \end{bmatrix}$$

$\text{Indep of } r_{12} \Rightarrow \vec{r}_1 + \vec{r}_2$
 $\Rightarrow$ Trivially satisfies C.C. Condition.

Let's denote $\rho(\vec{r}_i; i=1,2,\dots,N) = \rho(\vec{r}_{ij}; 1 \leq i < j \leq N) = \rho(r_{ij}; 1 \leq i < j \leq N)$

There are 2 invariant principles for ρ ;

$$\left\{ \begin{array}{l} \text{Scaling Invariance} \quad \rho(\alpha \vec{r}_i) = \rho(\vec{r}_i) \quad \forall \alpha > 0 \\ \text{Rotational " " : Define } \bar{T}_{3 \times 3} \text{ as a rotation operator.} \end{array} \right.$$

$$\bar{T}^T \cdot \bar{T} = \bar{I} \text{ unit Id}$$

e.g. $T = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $\rho(\bar{T} \cdot \vec{r}_i) = \rho(\vec{r}_i)$

$$|\bar{T} \cdot \vec{r}_{ij}| = \sqrt{\vec{r}_{ij} \cdot \underbrace{\bar{T}^T \cdot \bar{T}}_{\bar{I}} \cdot \vec{r}_{ij}} = \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}} = |\vec{r}_{ij}|$$

Let's define a C.C. $\vec{X}^* = \begin{bmatrix} \vec{r}_1^* \\ \vec{r}_2^* \\ \vdots \\ \vec{r}_N^* \end{bmatrix}$ $* \equiv \text{C.C.}$

s.t. $\frac{J\mathcal{L}}{J\vec{X}} = \vec{0}_{3N \times 1}$

Then $\vec{X}' = \alpha \vec{T}_{3N \times 3N} \cdot \vec{X}^*$ is a C.C. $\vec{T}_{3N \times 3N} = \begin{bmatrix} \vec{T} & & \\ & \vec{T} & \\ & & \ddots \\ & & & \vec{T} \end{bmatrix}$

where $\alpha + \vec{T}$ can be arbitrary functions of time.

Assume $\vec{T}$ corresponds to a rotation Θ about a fixed axis $\hat{\Theta}$.

" $\vec{X}^*$ lies in a single plane $\perp \hat{\Theta}$.

Thus $\dot{\vec{T}} = -\dot{\Theta} \hat{\Theta} \times \vec{T} = -\dot{\Theta} \hat{\Theta} \cdot \vec{T}$

Define the single body in our C.C.

$$\vec{r}_i' = \alpha \vec{T} \cdot \vec{r}_i^*$$

$$\dot{\vec{r}}_i' = \dot{\alpha} \vec{T} \cdot \vec{r}_i^* + \alpha \dot{\vec{T}} \cdot \vec{r}_i^* + \alpha \vec{T} \cdot \dot{\vec{r}}_i^* = \dot{\alpha} \vec{T} \cdot \vec{r}_i^* - \alpha \dot{\theta} \hat{\theta} \cdot \vec{T} \cdot \vec{r}_i^* + \alpha \vec{T} \cdot \dot{\vec{r}}_i^*$$

$$\ddot{\vec{r}}_i' = \ddot{\alpha} \vec{T} \cdot \vec{r}_i^* - 2\dot{\alpha}\dot{\theta} \hat{\theta} \cdot \vec{T} \cdot \vec{r}_i^* - \alpha \ddot{\theta} \hat{\theta} \cdot \vec{T} \cdot \vec{r}_i^* + \alpha \dot{\theta}^2 \hat{\theta} \cdot \hat{\theta} \cdot \vec{T} \cdot \vec{r}_i^* + \alpha \vec{T} \cdot \ddot{\vec{r}}_i^* = 2\sigma \vec{r}_i'$$

$i=1, 2, \dots, N$

Can decompose this into 2 directions in the plane $\perp \hat{\theta}$

$$\ddot{\vec{r}}_i' = [\ddot{\alpha} - \alpha \dot{\theta}^2] \vec{T} \cdot \vec{r}_i^* - [2\dot{\alpha}\dot{\theta} + \alpha \ddot{\theta}] \hat{\theta} \cdot \vec{T} \cdot \vec{r}_i^* = 2\sigma \alpha \vec{T} \cdot \vec{r}_i^*$$

$$\ddot{\alpha} - \alpha \dot{\theta}^2 = 2\sigma \alpha$$

$$2\dot{\alpha}\dot{\theta} + \alpha \ddot{\theta} = 0$$

where $\sigma = \frac{1}{2} \frac{v}{r_p}$

What is σ ?

$$\sigma = \frac{1}{2} \frac{\mathcal{U}}{I_p}$$

is not invariant w/ α
is invariant with $\bar{T}$

$$\sigma = \frac{1}{2} \frac{\mathcal{U}(\alpha \bar{T} \cdot \vec{r}_i^*)}{I_p(\alpha \bar{T} \cdot \vec{r}_i^*)} = \frac{1}{2} \frac{\mathcal{U}(\alpha \vec{r}_i^*)}{I_p(\alpha \vec{r}_i^*)} = \frac{1}{2} \frac{1}{\alpha} \mathcal{U}(\vec{r}_i^*) \cdot \frac{1}{\alpha^2 I_p(\vec{r}_i^*)}$$

$$\sigma = \frac{1}{2\alpha^3} \frac{\mathcal{U}(\vec{r}_i^*)}{I_p(\vec{r}_i^*)} = \frac{\sigma^*}{\alpha^3}$$

$$\ddot{\alpha} - \alpha \dot{\theta}^2 = \frac{2\alpha\sigma^*}{\alpha^3} = \frac{2\sigma^*}{\alpha^2} ; \quad 2\dot{\alpha}\dot{\theta} + \alpha\ddot{\theta} = 0$$

$$N=2 \quad \sigma^* = \frac{1}{2} \frac{\mathcal{U}(R^*)}{I_p(R^*)} = -\frac{GM_1M_2}{2R^*} \cdot \left(\frac{M_1M_2}{M_1+M_2} \right) R^{*2} = \left(\frac{-G(M_1+M_2)}{2R^{*3}} \right)$$

Solution: $\alpha(2\dot{\alpha}\dot{\theta} + \alpha\ddot{\theta}) = 0 \Rightarrow 2\alpha\dot{\alpha}\dot{\theta} + \alpha^2\ddot{\theta} = \frac{d}{dt}(\alpha^2\dot{\theta}) = 0$

$$\alpha^2\dot{\theta} = \text{constant} = \beta \Rightarrow \boxed{\dot{\theta} = \frac{\beta}{\alpha^2}}$$

$$\boxed{\ddot{\alpha} - \frac{\beta^2}{\alpha^3} = \frac{2\sigma^*}{\alpha^2}}$$

Let $\alpha = \frac{1}{u}$

time $\rightarrow \theta$

$$\left(\frac{d}{dt} = \dot{\theta} \frac{d}{d\theta} = \frac{\beta}{\alpha^2} \frac{d}{d\theta} = u^2 \beta \frac{d}{d\theta} \right)$$

$$\ddot{\alpha} = -\beta^2 u'' u^2 \quad \left(\frac{d}{d\theta} u = u' \right)$$

$$-\beta^2 u^2 u'' - u^3 \beta^2 = 2\sigma^* u^2 \Rightarrow$$

$$\boxed{u'' + u = \frac{-2\sigma^*}{\beta^2}}$$

Solution

$$\boxed{u(\theta) = A \cos \theta + B \sin \theta - \frac{2\sigma^*}{\beta^2}}$$

Choose IC's at $\alpha(0) = \alpha_0$, minimum

$$\alpha(0) = A - \frac{2\sigma^4}{\beta^2} = \frac{1}{\alpha_0} \quad A = \frac{1}{\alpha_0} + \frac{2\sigma^4}{\beta^2}$$

$$\alpha'(0) = \boxed{0 = \beta}$$



p

$$\alpha(\theta) = \frac{-\frac{\beta^2}{2\sigma^4}}{1 - \underbrace{\left(1 + \frac{\beta^2}{2\sigma^4\alpha_0}\right)}_e \cos\theta}$$

$$\alpha(\theta) = \frac{p}{1 + e \cos\theta}$$

$$p = \frac{-\beta^2}{2\sigma^4}$$

$$e = -\left[\frac{\beta^2}{2\sigma^4\alpha_0} + 1\right]$$

Kepler's Eqn.

$$\frac{d\theta}{dt} = \frac{\beta}{\alpha^2(\theta)}$$

$$\int_0^\theta \alpha^2(\theta) d\theta = \int_{t_0}^t \beta dt = \beta(t - t_0)$$

$$\text{For } N=2$$

$$\sigma^* = \frac{-G(m_1+m_2)}{2 r_*^3}$$

$$p = \frac{\beta^2 r_*^3}{G(m_1+m_2)}$$

$$e = \frac{\beta^2 r_*^3}{G(m_1+m_2) \alpha_0} - 1$$

Alternate approach, only finds relative equilibria ($e=0$). Recall the Amended Potential (Planar)

$$\mathcal{E}_p = \frac{H^2}{2I_p} + \mathcal{U}$$

$$\Rightarrow \frac{J \mathcal{E}_p}{J \vec{R}} = \vec{0} \quad \begin{matrix} 3N \times 1 \\ 3N \times 1 \end{matrix}$$

defines a
rel. Equilibrium

Thm: IF $\left. \frac{J\mathcal{U}}{J\vec{R}} \right|_{\vec{R}^*} = \vec{0}$ and all the $\vec{R}^*$ are coplanar,

then there exists a H value such that $\left. \frac{J\mathcal{E}_p}{J\vec{R}} \right|_{\vec{R}^*} = \vec{0}$

Proof: $\frac{J\mathcal{E}}{J\vec{R}} = 2\mathcal{U}I_p \left. \frac{J\mathcal{U}}{J\vec{R}} \right|_* + \mathcal{U}^2 \left. \frac{JI_p}{J\vec{R}} \right|_* = 0$

So $\left. \frac{J\mathcal{U}}{J\vec{R}} \right|_* = -\frac{\mathcal{U}}{2I_p} \left. \frac{JI_p}{J\vec{R}} \right|_*$

Consider $\mathcal{E}_p$ w/ constant A.M.

$$\frac{J\mathcal{E}_p}{J\vec{R}} = \frac{-H^2}{2I_p^2} \left. \frac{JI_p}{J\vec{R}} \right|_* + \left. \frac{J\mathcal{U}}{J\vec{R}} \right|_* = \frac{-1}{2I_p^2} [H^2 + I_p \mathcal{U}] \left. \frac{JI_p}{J\vec{R}} \right|_*$$

Implications

- If we find a C.C. we can arbitrarily scale it by the "2BP" solution (conic sections)
- The "2BP" solution is just a special case of C.C. motion.

A Homographic Solution is a dynamically evolving system that remains in a C.C. for all time, must follow the 2BP.

- A Homothetic Solution is a Homographic Solution w/ $H = 0$.

- A non-planar homographic solution must be a homothetic solution.

A total collapse of N bodies ($H = 0$) will asymptotically approach a C.C. as $t \rightarrow t_c$.

Since $\mathcal{U} < 0$, $I_p > 0$, can choose

$$\boxed{H^2 = -I_p \mathcal{U}} \Rightarrow \text{yield eq. solution}$$

$I_p \mathcal{U}$ is homog. of degree 1

$$\boxed{H^2 = -\alpha I_p^* \mathcal{U}^*}$$

$\Omega = \text{ang. vel.}$

At a rel. Eq.

$$H^2 = (I_p \Omega)^2 = -\mathcal{U} I_p$$

$$\Omega^2 = \frac{-\mathcal{U}}{I_p} \Rightarrow \text{Take } e=0$$

$$\beta^2 = -2\sigma^* \alpha_0 = \alpha_0^4 \Omega^2$$

$$\Omega^2 = \frac{-|\mathcal{U}^*|}{\alpha_0^3 I_p^*} = \frac{-\mathcal{U}(\alpha, k^*)}{I_p(\alpha, R^*)}$$

□